

Sebastian Dziadzio

- AI researcher specializing in continual pretraining, and benchmarking open-ended and agentic systems.
- Ten years of machine learning research experience across startups, industry labs, and academia.
- Published at NeurIPS, CVPR, ACL, and ICCV, with 780+ citations and two patents.

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🏢 EXPERIENCE

- Feb 2022–Present • **Doctoral Researcher, Bethge Lab, Tübingen AI Center** · Tübingen, Germany
- Developed methods for continual pretraining, open-ended evaluation, inferring mechanisms from behavior, and scaling synthetic reinforcement learning environments.
- Built ONEBench, enabling robust model ranking with over 95% of scores missing, and RevengeBench, testing 12 frontier LLM agents on inferring opponent strategies as code.
- Oct 2019–Dec 2021 • **Scientist, Microsoft Research, Mixed Reality & AI Lab** · Cambridge, UK
- Owned a parametric 3D face model used by 30+ researchers across multiple teams.
- Rebuilt the training, evaluation, and visualization infrastructure, and led a research effort that reduced the average fitting error by half.
- Sep 2018–Sep 2019 • **AI Resident, Microsoft Research** · Cambridge, UK
- Built a real-time deep learning system for novel view synthesis, demoed at MSR TechFest.
- Initiated Project Paidia, integrating MSR's RL library into Ninja Theory's upcoming title, and implemented a reverse curriculum that doubled sample efficiency over PPO.
- Sep 2016–Aug 2018 • **Data Scientist, Cliqz** · Munich, Germany
- Owned the product telemetry pipeline. Identified an opportunity to improve web index retrieval, secured leadership backing, and built the initial deep learning model.
- Mar 2015–Jun 2016 • **Software Engineer, Nokia Networks** · Kraków, Poland
- Implemented LTE control plane features and designed system and component tests.

🎓 EDUCATION

- Feb 2022–Dec 2026 • **PhD, Computer Science, University of Tübingen**
ELLIS PhD Program, International Max Planck Research School for Intelligent Systems
- Mar 2014–Sep 2016 • **MSc, Computer Science, AGH University of Kraków**
- Oct 2010–Feb 2014 • **BSc, Acoustical Engineering, AGH University of Kraków**

⚙️ SKILLS

Research: continual multimodal pretraining, model merging, agentic systems, benchmarking, post-training

Research engineering: data curation, distributed training and inference, evaluation infrastructure

Tools: Python, PyTorch, DeepSpeed, FSDP, vLLM, Ray, Slurm, Docker, Hugging Face, Weights & Biases

Languages: Polish (native), English (fluent), German (intermediate), Spanish (conversational)

ACADEMIC LEADERSHIP

Organization: ML in PL Conference 2022–2026, team lead 2025, 700+ attendees; ML Summer School 2022–2026

Collaboration: initiated a joint project with the IDEAS Research Institute and KU Leuven

Supervision: led programming courses for marginalized communities at ReDI, supervised MSc and PhD students

Invited talks: Data Science Summit 2023, CVPR Workshop on Test-Time Adaptation 2024

Reviewing: NeurIPS, ICML, ICLR, CVPR

PUBLICATIONS

* denotes equal first-author contribution · Full list on [Google Scholar](#) 

RevengeBench: Reverse Engineering Code-Space Policies from Behavioral Experiments 

[NeurIPS 2026](#)

B. Rahmani*, S. Dziadzio*, J. Strüber*, S. Hernández-Gutiérrez, M. Bethge.

ONEBench to Test Them All: Sample-Level Benchmarking Over Open-Ended Capabilities 

[ACL 2025](#)

A. Ghosh*, S. Dziadzio*, A. Prabhu, V. Udandarao, S. Albanie, M. Bethge.

How to Merge Your Multimodal Models Over Time? 

[CVPR 2025](#)

S. Dziadzio*, V. Udandarao*, K. Roth*, A. Prabhu, Z. Akata, S. Albanie, M. Bethge.

A Practitioner's Guide to Continual Multimodal Pretraining 

[NeurIPS 2024](#)

K. Roth*, V. Udandarao*, S. Dziadzio, A. Prabhu, M. Cherti, O. Vinyals, O. Hénaff, S. Albanie, M. Bethge, Z. Akata.

Disentangled Continual Learning: Separating Memory Edits from Generalization 

[CoLLAs 2024](#)

S. Dziadzio, Ç. Yıldız, G. M. van de Ven, T. Trzciński, T. Tuytelaars, M. Bethge.

Full-Body Motion from a Single Head-Mounted Device 

[ICCV 2021](#)

A. Dittadi, S. Dziadzio, D. Cosker, B. Lundell, T. Cashman, J. Shotton.

Fake It Till You Make It: Face Analysis in the Wild Using Synthetic Data Alone 

[ICCV 2021](#)

E. Wood, T. Baltrušaitis, C. Hewitt, S. Dziadzio, T. Cashman, J. Shotton.

PATENTS

Controllable Image Generation 

[US-11748932, 2023](#)

M. Kowalski, S. Garbin, M. Johnson, T. Baltrušaitis, M. De La Gorce, V. Estellers, S. Dziadzio.

Computing Photorealistic Versions of Synthetic Images 

[US-11354846, 2022](#)

S. Garbin, M. Kowalski, M. Johnson, T. Baltrušaitis, M. De La Gorce, V. Estellers, S. Dziadzio, J. Shotton.